

Data Dictionary

Ordinal-Probit Latent-Class Trajectory Model

File: ordinal_probit.sas

Last updated: July 21, 2026

Package: Traj2 (released core)

1. Overview

This file fits a single-outcome, class-specific cumulative-probit trajectory model for bounded ordinal outcomes (e.g. ADL, frailty, cognitive scores), by maximum likelihood in PROC NLMIXED.

Class-specific thresholds are built from exponential increments to guarantee ordering without constrained optimization.

2. Datasets

Name	Description
sim_long / sim_wide	Long / wide outputs of the optional simulator (%sim_data).
BASE_FILE_SRS	Wide input table with BENE_ID, quar1..quarT (values 1..T), and the ordinal outcome columns.
&prefix._EST_K&k	Parameter estimates for a K-class fit.
&prefix._FIT_K&k	Fit statistics (AIC, BIC, -2 log L).
&prefix._ESTS_K&k	Additional estimates, natural-scale thresholds.
mix_props	Estimated mixing proportions for the class-proportion plot.
pred_long	Predicted $E[Y_t \text{class}]$ over time for the trajectory plot.

3. Macro Arguments (%ordprob_mix_fit_one)

Name	Type / Default	Description
data	name, BASE_FILE_SRS	Input wide-format dataset.
id	name, BENE_ID	Subject identifier.
yvars	varlist, required	Repeated-measure outcome vars (e.g. Y1_1-Y1_12).
tvars	varlist, required	Time-index vars (e.g. quar1-quar12).
ttotal	integer, 12	Number of time points.
m	integer, 4	Number of ordinal categories.
ycodes	list, 0 1 2 3	Category codes.

Name	Type / Default	Description
deg	integer, 2	Polynomial degree of class trajectories.
k	integer, 4	Number of latent classes (C).
qpoints	integer, 40	Quadrature points.
maxiter	integer, 1000	Maximum optimizer iterations.
tech	name, dbldog	PROC NL MIXED optimization technique.
start_values	text, optional	User starting values (increments only; blank = 0).
bounds	text, optional	Parameter bounds for stability.
outestlib	libref, work	Library for output datasets.
prefix	name, ordprob_single	Prefix for output dataset names.

4. Key Variables and Parameters

Name	Type	Description
BENE_ID	Num	Subject identifier.
yv[ttotal]	Array	Ordinal outcome over time inside the model.
tv[ttotal]	Array	Time predictor inside the model.
cat	Num	Category index (1..m) matched from ycodes.
alpha0_[class]	Num	Class-membership logit (class A fixed to 0).
beta_[class][k]	Num	Degree-k polynomial coefficient of the latent index eta.
eta_[class]	Num	Class-specific latent index at time t.
i[class]1	Num	First threshold parameter (fixed at 0 for identification).
i[class]j (j>=2)	Num	Exp-increment producing threshold j.
th[j]_[class]	Num	Cumulative threshold j, ordered by construction.
z[j]_[class]	Num	Category-j probability from probit differences.
pie[c]	Num	Marginal class-membership probability (softmax).
llik[c]	Num	Per-class log-likelihood accumulator.
mix	Num	Mixture likelihood passed to the model statement.

5. Macros

Name	Purpose
%sim_data	Optional simulator: BIN4, TPOIS4, or CAT5 ordinal DGP.
%build_base_from_simwide	Builds BASE_FILE_SRS with quar1..quarT = 1..T.
%CL	Maps a class index to its letter label (A..O).
%index_override_pairs	Flags user-supplied starts so defaults skip them.
%make_default_parms	Builds default zero starts, skipping overrides.
%ordprob_mix_fit_one	Fits the ordinal-probit model (single PARMS statement).
%ordprob_mix_plot_one	Class-proportion barplot and predicted-trajectory curves.

6. Notes and Assumptions

- Missing time points are allowed and skipped in the likelihood (missing at random).
- Identification: the first threshold is fixed at 0 with beta_[class]0 free; supply only increments i[class]2..i[class](m-1) in start_values, never the first threshold and never the class A mixing logit.
- Exp-increment thresholds guarantee ordered categories without bounds.
- cap in %sim_data sets the highest category index; cap=3 yields values {0,1,2,3}, matching m=4 and ycodes=0 1 2 3.
- Fit statistics across candidate k values support model selection by BIC.